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AUTOMATIC IMAGE CAPTION GENERATOR

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (A)

(Affiliated to Osmania University)

CASE STUDY REPORT

Submitted By:

160124748094 – A.Hemanth

160124748121 – M.Dhanush

CASE STUDY GUIDE

Panigrahi Srikanth, Assistant Professor

Department of CSE Artificial Intelligence and Machine Learning

CBIT, Hyderabad

Certificate

This is to certify that the project report entitled **"Automatic Image Caption Generator"** has been carried out by the following students during the academic year 2026–2027 under the guidance and supervision of the project guide.

160124748094 – A.Hemanth
160124748121 – M.Dhanush

Project Guide:

Panigrahi Srikanth

Assistant Professor

Department of CSE AI&ML

CBIT, Hyderabad

Acknowledgement

We express our sincere gratitude to everyone who supported us throughout the development of this case study. We thank our guide, **Panigrahi Srikanth**, for his continuous guidance, encouragement and valuable suggestions throughout the project.

We also thank the Department of CSE Artificial Intelligence and Machine Learning, CBIT, Hyderabad, for providing the required academic environment and computational resources.

Finally, we express our gratitude to our family members, friends and classmates for their support and encouragement.

Abstract

Automatic image caption generation is a deep learning task that combines computer vision and natural language processing to generate a meaningful textual description for an input image. The objective of this case study is to develop an image captioning system using the Flickr8k dataset.

The proposed system uses a pretrained VGG16 convolutional neural network as the image encoder. The extracted spatial image features are processed using a Bahdanau attention mechanism, which allows the decoder to focus on relevant regions of the image while generating each word. An LSTM network is used as the sequence decoder.

The model incorporates several deep learning concepts including activation functions, softmax, cross-entropy, Adam optimization, gradient clipping, batch normalization, dropout, L2 regularization and early stopping. Beam Search is used during inference to improve the quality of generated captions.

The system was evaluated using BLEU scores. On a sample of 10 unseen Flickr8k test images, the model obtained BLEU-1, BLEU-2, BLEU-3 and BLEU-4 scores of 0.6265, 0.3826, 0.2338 and 0.1675 respectively. The results demonstrate the ability of the proposed system to learn relationships between visual features and natural language captions.

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1. Introduction

1.1 Background

Image caption generation is the task of automatically producing a natural-language description of the content of an image. It combines computer vision, deep learning and natural language processing.

Traditional computer vision systems generally identify objects or classify images, whereas image captioning attempts to understand the relationship between multiple visual elements and express the information as a sentence.

Deep learning has enabled significant progress in this area through the combination of convolutional neural networks for visual feature extraction and recurrent neural networks for language generation.

1.2 Problem Statement

The objective of this case study is to develop a deep learning system that accepts an image as input and automatically generates a meaningful natural-language caption describing the image.

The system should:

- Extract meaningful visual features from an image.
- Represent the image using spatial feature representations.
- Generate captions word by word.
- Use attention to focus on relevant image regions.
- Evaluate generated captions using standard metrics.

1.3 Objectives

The main objectives are:

1. To understand the fundamentals of deep learning based image caption generation.
2. To extract image features using a pretrained VGG16 network.
3. To implement an LSTM based language decoder.
4. To implement Bahdanau attention.
5. To study optimization techniques such as SGD, Momentum, RMSProp and Adam.
6. To apply dropout, L2 regularization, batch normalization and gradient clipping.
7. To evaluate generated captions using BLEU scores.

1.4 Scope

The developed system is intended as an academic deep learning case study. It demonstrates the complete pipeline from image feature extraction to natural-language caption generation.

The system is trained using Flickr8k and therefore its ability to describe arbitrary images is limited by the visual and linguistic coverage of the training dataset.

2. Literature Review and Theoretical Study

2.1 Deep Learning for Image Captioning

Image captioning generally consists of two major components: an image encoder and a language decoder.

The encoder transforms an image into a numerical feature representation, while the decoder generates a sequence of words conditioned on those features.

2.2 Artificial Neuron and MLP

An artificial neuron receives input values, applies weights and bias, and passes the result through an activation function.

$$z = \sum_{i=1} w_i x_i + b$$

The output is:

$$y = f(z)$$

Multiple neurons arranged in layers form a multilayer perceptron.

2.3 Activation Functions

Common activation functions include ReLU, sigmoid and softmax.

ReLU is defined as:

$$f(x) = \max(0, x)$$

The softmax function converts the final decoder outputs into a probability distribution over the vocabulary:

$$P(y_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

2.4 Cross-Entropy Loss

The model uses categorical prediction at each time step. Sparse categorical cross-entropy is used to measure the difference between the target word and the predicted word distribution.

2.5 Convolutional Neural Networks

CNNs are widely used for image feature extraction. A convolutional layer applies learnable filters to an image to detect visual patterns such as edges, textures and higher-level structures.

2.6 VGG16

VGG16 is a pretrained convolutional neural network trained on ImageNet. In this project, the classification layers are removed using:

```
VGG16(weights="imagenet", include_top=False)
```

For an input image, the resulting feature representation has the shape:

$$7 \times 7 \times 512$$

The spatial dimensions are reshaped into:

$$49 \times 512$$

where the 49 locations represent spatial image regions.

2.7 Recurrent Neural Networks

RNNs process sequential data and maintain information from previous time steps. They are suitable for language generation because a caption is a sequence of words.

2.8 LSTM

Long Short-Term Memory networks overcome some limitations of ordinary RNNs by using gates to control information flow.

The main gates are:

- Forget gate
- Input gate
- Output gate

The LSTM decoder generates the caption sequentially.

2.9 GRU and Bidirectional LSTM

GRU is an alternative recurrent architecture with fewer gates than LSTM. Bidirectional LSTM processes sequences in both directions.

These architectures were studied as part of the theoretical comparison, while the implemented caption generator uses LSTM.

2.10 Attention Mechanism

Attention allows the decoder to assign different importance to different spatial regions of the image.

The implemented system uses Bahdanau attention.

The attention score can be represented as:

$$e_{t,i} = V^T \tanh(W_1 h_i + W_2 s_t)$$

The attention weights are obtained using softmax:

$$\alpha_{t,i} = \frac{\exp(e_{t,i})}{\sum_j \exp(e_{t,j})}$$

The context vector is:

$$c_t = \sum_i \alpha_{t,i} h_i$$

This context is combined with the LSTM output to predict the next word.

2.11 Optimization Techniques

The project incorporates the following optimization and regularization concepts:

- Stochastic Gradient Descent
- Momentum
- RMSProp
- Adam
- Gradient clipping
- Batch normalization
- Dropout
- L2 regularization
- Early stopping

2.12 BLEU Metric

BLEU is used to compare generated captions with reference captions using n-gram overlap.

BLEU-1 considers individual words, while BLEU-2, BLEU-3 and BLEU-4 consider increasingly longer word sequences.

3. System Requirements and Specifications

3.1 Hardware Requirements

The experiment was developed using Google Colab with GPU acceleration.

Table 3.1: Hardware and computational environment

Component	Specification
Execution Platform	Google Colab
GPU	NVIDIA T4
Dataset	Flickr8k
Image Input Size	224×224
CNN Feature Size	$7 \times 7 \times 512$

3.2 Software Requirements

Table 3.2: Software and libraries

Software/Library	Purpose
Python	Programming language
TensorFlow/Keras	Deep learning framework
NumPy	Numerical processing
Pandas	Dataset handling
Matplotlib	Visualization
NLTK	BLEU evaluation
Pillow	Image processing
Kaggle API	Dataset download

3.3 Dataset Description

The Flickr8k dataset is used for training and evaluation. It contains images paired with natural-language captions.

The dataset is divided into training, validation and testing subsets using unique images to avoid image-level leakage between the splits.

3.4 Data Preprocessing

The captions are converted to lowercase and punctuation is removed. Special tokens are added:

```
startseq
...
endseq
```

The training captions are tokenized and converted into integer sequences.

Images are resized to 224×224 before being passed through VGG16.

4. Methodology and System Design

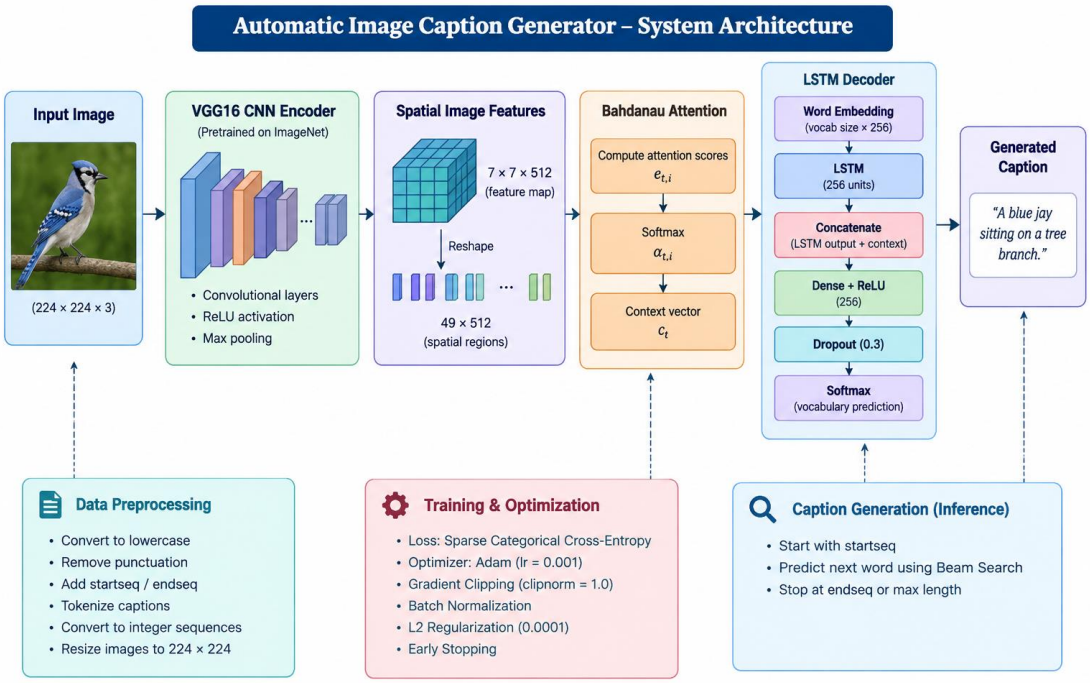
4.1 Overall Workflow

The complete workflow is:

1. Load the Flickr8k dataset.
2. Clean and tokenize captions.
3. Split images into training, validation and test sets.
4. Extract image features using VGG16.
5. Convert the feature map into 49 spatial regions.
6. Process caption sequences using an embedding layer and LSTM.
7. Apply Bahdanau attention over image regions.
8. Combine attention context with LSTM output.
9. Predict the next word using softmax.
10. Train using Adam and masked cross-entropy loss.
11. Generate captions using Beam Search.
12. Evaluate using BLEU scores.

4.2 System Architecture

The proposed architecture is:



4.3 Image Encoder

The VGG16 model is used without its final classification layers. The output feature map contains 49 spatial regions, each represented by 512 features.

4.4 Caption Decoder

The decoder uses an embedding layer followed by an LSTM with 256 hidden units.

4.5 Attention-Based Decoding

At each time step, attention computes the relevance of the 49 image regions with respect to the current language representation.

The resulting context vector is concatenated with the LSTM output before the final vocabulary prediction layer.

4.6 Regularization

The model uses:

- Dropout with probability 0.3.
- L2 regularization with coefficient 0.0001.
- Batch normalization.
- Gradient clipping using `clipnorm=1.0`.
- Early stopping.

4.7 Caption Generation

During inference, the model starts with `startseq` and predicts one word at a time until `endseq` is generated or the maximum caption length is reached.

Beam Search is used instead of simple greedy decoding to retain multiple candidate sequences during generation.

5. Implementation Details and Experimental Setup

5.1 Implementation Environment

The project was implemented using Python and TensorFlow/Keras in Google Colab with NVIDIA T4 GPU acceleration.

5.2 Dataset Loading

The Flickr8k dataset and caption file were downloaded using the Kaggle API.

5.3 Caption Cleaning

The following operations were performed:

- Convert captions to lowercase.
- Remove non-alphabetic characters.
- Remove unnecessary whitespace.
- Add `startseq` and `endseq`.

5.4 Feature Extraction

VGG16 was used to extract spatial image features.

The feature representation is:

$$7 \times 7 \times 512$$

and is reshaped to:

$$49 \times 512$$

for the attention mechanism.

5.5 Training Configuration

Table 5.1: Main model configuration

Parameter	Value
CNN Encoder	VGG16
Feature Size	49×512
Embedding Dimension	256
LSTM Units	256
Attention Units	256
Dropout	0.3
L2 Regularization	0.0001
Optimizer	Adam
Initial Learning Rate	0.001
Gradient Clipping	1.0
Batch Size	64

5.6 Training Output

```
Epoch 1/5
506/506 ━━━━━━━━━━━━━━━━━━━━ 46s 76ms/step - loss: 4.4180 - masked_accuracy: 0.2675 - val_loss: 3.6479 - val_masked_accuracy: 0.3411 - learning_rate: 0.0010
Epoch 2/5
506/506 ━━━━━━━━━━━━━━━━━━━━ 38s 75ms/step - loss: 3.4139 - masked_accuracy: 0.3588 - val_loss: 3.3545 - val_masked_accuracy: 0.3741 - learning_rate: 0.0010
Epoch 3/5
506/506 ━━━━━━━━━━━━━━━━━━━━ 38s 75ms/step - loss: 3.0824 - masked_accuracy: 0.3870 - val_loss: 3.2372 - val_masked_accuracy: 0.3906 - learning_rate: 0.0010
Epoch 4/5
506/506 ━━━━━━━━━━━━━━━━━━━━ 38s 75ms/step - loss: 2.8721 - masked_accuracy: 0.4075 - val_loss: 3.1985 - val_masked_accuracy: 0.3948 - learning_rate: 0.0010
Epoch 5/5
506/506 ━━━━━━━━━━━━━━━━━━━━ 38s 75ms/step - loss: 2.7080 - masked_accuracy: 0.4247 - val_loss: 3.2010 - val_masked_accuracy: 0.4012 - learning_rate: 0.0010
```

Figure 5.1: Training output of the image caption generation model.

5.7 Training and Validation Loss

```
64/64 ━━━━━━━━━━━━━━━━━━━━ 2s 38ms/step - loss: 3.2520 - masked_accuracy: 0.3874
Test Loss: 3.2519731521606445
Test Accuracy: 0.3874336481094360
```

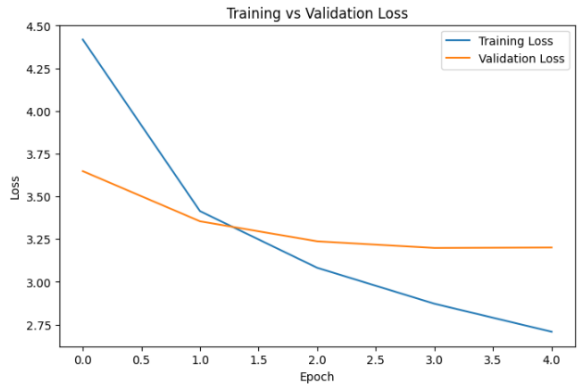


Figure 5.2: Training and validation loss.

5.8 Training and Validation Accuracy

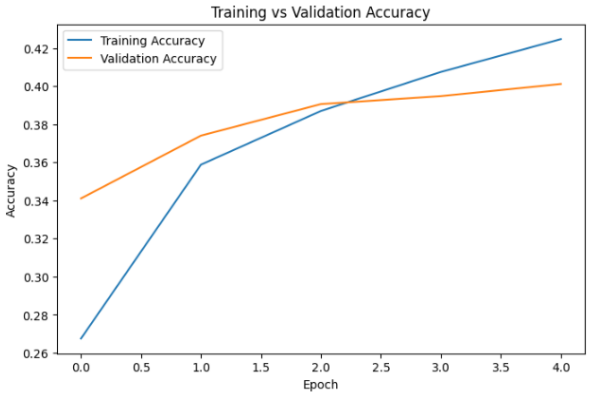


Figure 5.3: Training and validation masked accuracy.

6. Results and Performance Analysis

6.1 Overview

The trained model was evaluated using generated captions on unseen Flickr8k test images. BLEU metrics were used to measure the overlap between generated captions and reference captions.

6.2 Quantitative Performance

The following scores were obtained using Beam Search on a sample of 10 unseen Flickr8k test images.

Table 6.1: BLEU evaluation results

Metric	Score
BLEU-1	0.6265
BLEU-2	0.3826
BLEU-3	0.2338
BLEU-4	0.1675

6.3 BLEU Score Visualization

BLEU-1 0.6265
BLEU-2 0.3826
BLEU-3 0.2338
BLEU-4 0.1675

Figure 6.1: BLEU score values obtained during evaluation.

6.4 Interpretation of BLEU Results

BLEU-1 achieved the highest value because unigram matching is less restrictive and measures individual word overlap.

As the BLEU order increases, increasingly longer word sequences must match the reference captions. Therefore, BLEU-2, BLEU-3 and BLEU-4 are progressively lower.

The obtained results indicate that the model is able to generate captions containing meaningful words and phrases that overlap with human-written reference captions.

It should be noted that the reported scores were calculated using 10 unseen test images and therefore represent a sample evaluation rather than a complete benchmark over the entire test set.

6.5 Qualitative Analysis

The generated captions were visually inspected by comparing them with the corresponding input images and reference captions.



Figure 6.2: Sample image used for caption generation.

Generated Caption:

Generated Caption:
a black dog is running through the snow

Reference Caption:

Reference Caption:
a black dog running through the snow



Figure 6.3: Second sample image and generated caption.

Generated Caption:

Generated Caption:
a surfer is surfing on a wave

Reference Caption:

Reference Caption:
a lone surfboarder on a white surfboard flying through the air over a wave



Figure 6.4: Third sample image and generated caption.

Generated Caption:

Generated Caption:
three young girls are standing in front of a building

Reference Caption:

Reference Caption:
a girl with a white backpack is standing and smaller children are sitting in a row on the ground

6.6 Sample Implementation Output

The following output demonstrates the generation of a caption for an unseen image.



6.7 Optimizer Comparison

The implementation includes SGD, Momentum, RMSProp and Adam as part of the optimization study. Adam was selected as the primary optimizer for the final model because it provides adaptive learning rates and performed effectively during training.

```
Training with: SGD
506/506 ━━━━━━━━━━━━━━━━━━━━ 39s 73ms/step - loss: 2.6648 - masked_accuracy: 0.4286 - val_loss: 3.1948 - val_masked_accuracy: 0.3994

Training with: Momentum
506/506 ━━━━━━━━━━━━━━━━━━━━ 40s 73ms/step - loss: 2.6389 - masked_accuracy: 0.4319 - val_loss: 3.1994 - val_masked_accuracy: 0.4001

Training with: RMSprop
506/506 ━━━━━━━━━━━━━━━━━━━━ 41s 75ms/step - loss: 2.6814 - masked_accuracy: 0.4263 - val_loss: 3.1954 - val_masked_accuracy: 0.3962

Training with: Adam
506/506 ━━━━━━━━━━━━━━━━━━━━ 42s 76ms/step - loss: 2.8196 - masked_accuracy: 0.4242 - val_loss: 3.2099 - val_masked_accuracy: 0.3956

Optimizer Comparison
          Training Loss  Validation Loss  Training Accuracy  \
SGD            2.664762         3.194810           0.428569
Momentum       2.638899         3.199439           0.431945
RMSprop        2.681399         3.195401           0.426341
Adam           2.819553         3.209902           0.424157

          Validation Accuracy
SGD                  0.399359
Momentum             0.400066
RMSprop              0.396185
Adam                 0.395636
```

Figure 6.5: Optimizer comparison output.

6.8 Discussion

The results demonstrate that the proposed VGG16–LSTM architecture with Bahdanau attention can learn a relationship between image features and natural-language captions.

The attention mechanism provides a way to associate different words with different spatial regions of the image. Beam Search further improves the generation process by maintaining multiple candidate caption sequences.

However, the model is trained on Flickr8k, which is relatively small compared with modern image-language datasets. Therefore, performance on arbitrary Internet images may be limited, particularly for objects or scenes that are poorly represented in the training dataset.

7. Conclusion and Future Scope

7.1 Conclusion

This case study implemented an end-to-end automatic image caption generation system using deep learning.

The proposed system combines a pretrained VGG16 CNN encoder with an LSTM language decoder and Bahdanau attention. The system also demonstrates several important deep learning concepts including softmax classification, cross-entropy loss, Adam optimization, gradient clipping, batch normalization, dropout, L2 regularization and early stopping.

Beam Search was used during inference to generate improved caption sequences.

The model achieved BLEU-1, BLEU-2, BLEU-3 and BLEU-4 scores of 0.6265, 0.3826, 0.2338 and 0.1675 respectively on a sample of 10 unseen Flickr8k test images.

7.2 Limitations

The main limitations are:

- Flickr8k is relatively small for a general-purpose image captioning system.
- The model may generate incorrect captions for images that are very different from the training distribution.
- BLEU scores depend on the reference captions available in the dataset.
- The current evaluation uses a sample of 10 test images.
- The model does not provide guaranteed object-level recognition for arbitrary images.

7.3 Future Scope

Future improvements may include:

- Training on larger datasets such as MS COCO.
- Using stronger CNN or vision-transformer encoders.
- Fine-tuning the visual encoder.
- Using Transformer-based language decoders.
- Using more advanced image-language foundation models.
- Improving beam-search and decoding strategies.
- Evaluating with additional metrics such as METEOR, ROUGE-L and CIDEr.
- Developing a web-based interface for real-time caption generation.

8. Appendix I: Model Details and Preprocessing

8.1 Model Architecture

The implemented model consists of:

1. VGG16 pretrained CNN encoder.
2. Dense projection layer.
3. Batch normalization.
4. Word embedding layer.
5. LSTM decoder.
6. Bahdanau attention.
7. Concatenation of decoder and attention features.
8. Dense layer with ReLU activation.
9. Dropout.
10. Final softmax vocabulary layer.

8.2 Preprocessing

The image preprocessing pipeline consists of:

1. Load image.
2. Convert to RGB.

3. Resize to 224×224.
4. Apply VGG16 preprocessing.
5. Extract convolutional features.
6. Reshape $7 \times 7 \times 512$ into 49×512.

8.3 Caption Preprocessing

Caption preprocessing consists of:

1. Convert text to lowercase.
2. Remove punctuation and unwanted characters.
3. Add `startseq`.
4. Add `endseq`.
5. Tokenize the vocabulary.
6. Convert words to integer sequences.
7. Pad sequences to the maximum caption length.

8.4 Important Code Snippets

8.4.1 VGG16 Feature Extraction

```
vgg = VGG16(
    weights="imagenet",
    include_top=False
)
```

8.4.2 LSTM Decoder

```
lstm_output = layers.LSTM(
    256,
    return_sequences=True,
    dropout=0.3
)(word_embedding)
```

8.4.3 Adam Optimizer

```
optimizer = tf.keras.optimizers.Adam(
    learning_rate=0.001,
    clipnorm=1.0
)
```

8.4.4 Beam Search

The final caption generation stage uses Beam Search to maintain multiple candidate sequences rather than selecting only one word at each step.

8.5 Additional Code Output




```
BLEU-1: 0.5156629281406259
BLEU-2: 0.32726260069282714
BLEU-3: 0.20215562661930891
BLEU-4: 0.13505007207020703
```

Figure 8.1: Additional implementation output.

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